Systemy inteligentne 2 - Projekt				
Projekt nr.5				
Data zajęć: 18.12.2025	**Data oddania:** 15.01.2026	**Autor:** Marcin Hajduk	**Grupa:** 3ID16A	**Stanowisko:** 13

Założenia dotyczące przygotowania danych:

1. Obrazy uczące: **5** obrazów (obrazy **czterech** liter oraz **jednej** cyfry przygotowane w trakcie realizacji Projektu 3 - obrazy będące ”idealnymi” reprezentantami poszczególnych znaków przygotowane w matrycach **8x10 pikseli i 16x20 pikseli**)

2. Obrazy testowe: obrazy uczące oraz różnego rodzaju ich deformacje, w tym
   - obrazy ”idealne” z nałożonym szumem o różnym poziomie,
   - różne **fragmenty** obrazów ”idealnych”,
   - wybrane - w różny sposób zdeformowane - obrazy ”idealne”.

3. Celem eksperymentów jest zbadanie wpływu 3 przedstawionych poniżej czynników (punkty a), b) i c)) na dokładność funkcjonowania pamięci autoasocjacyjnej zbudowanej z wykorzystaniem sieci neuronowej Hopfielda. Czynniki te to:

   1) techniki zapisu do pamięci (tryb uczenia sieci) dostępne są:

      a. reguła Hebba (RH),
      b. reguła pseudoinwersji (RP),
      c. reguła rzutowania delta (RRD),
      d. zmodyfikowana reguła perceptronu (ZRP),

   2) techniki odczytu pamięci (tryb odtwarzania) - dostępne są:

      a. metoda synchroniczna (MS),
      b. metoda asynchroniczna losowa (MAL),
      c. metoda asynchroniczna cykliczna (MAC).

   3) pojemności pamięci (można ją zwiększać zwiększając rozmiar matrycy; oprócz początkowego rozmiaru 8x10 pikseli oraz 16x20 pikseli).

Tabela 1. Zbiór obrazów uczących i testowych 8x10 pikseli (U-uczący, T-testowy)

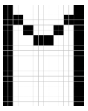
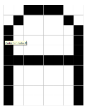
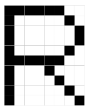
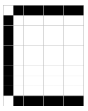
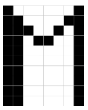
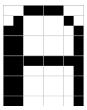
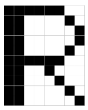
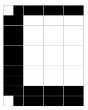
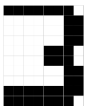
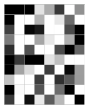
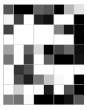
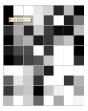
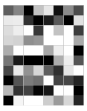
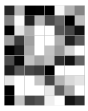
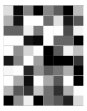
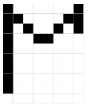
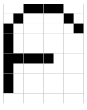
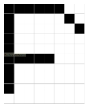
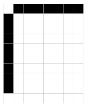
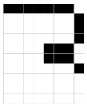
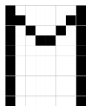
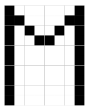
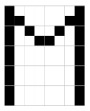
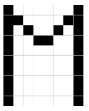
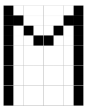
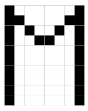
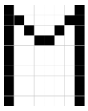
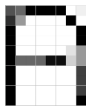
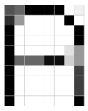
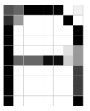
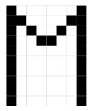
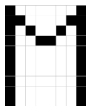
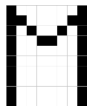
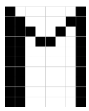
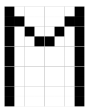
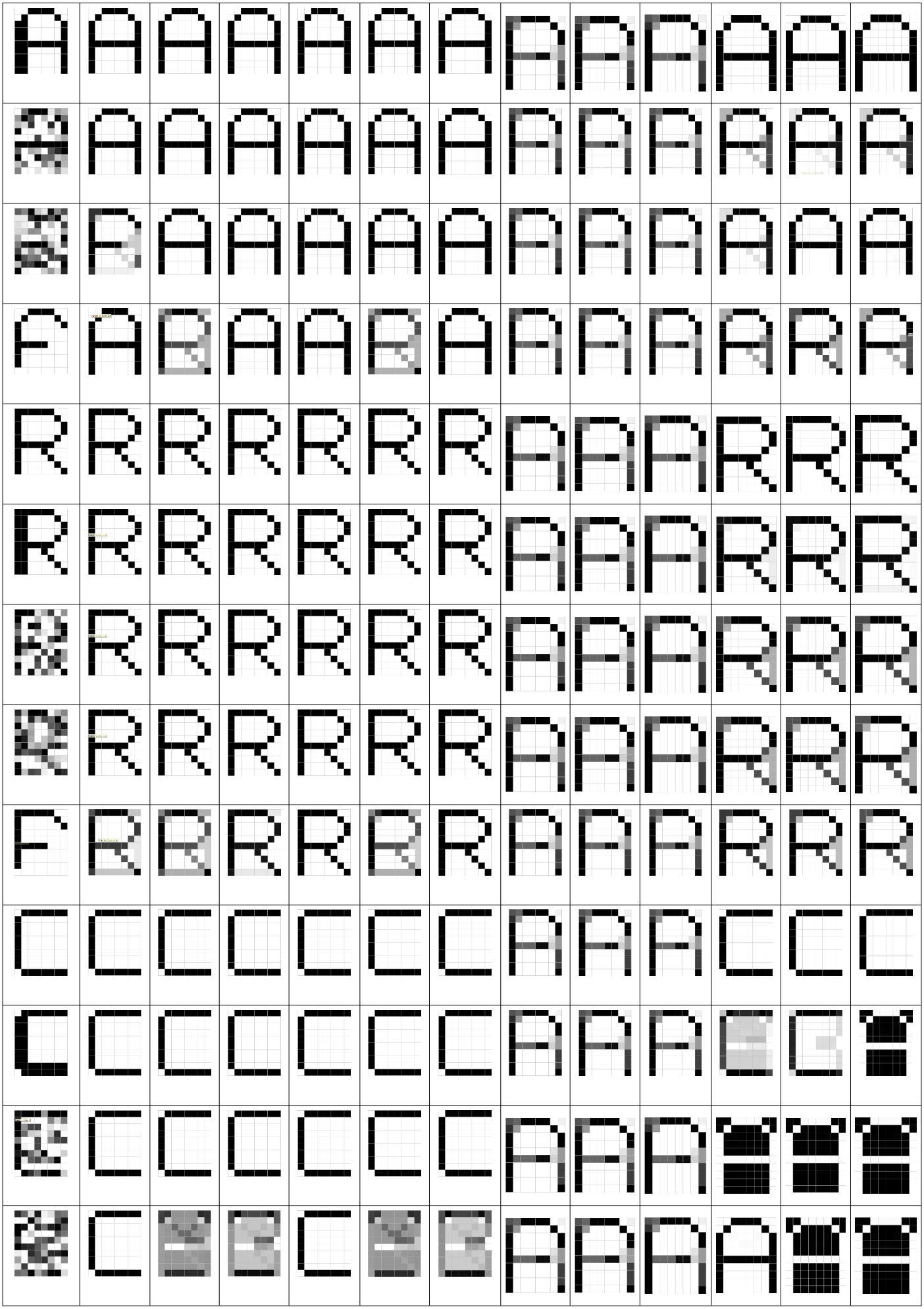
Litera	M	A	R	C	3
Obraz idealny (U)					
Zdeformowany obraz idealny (T)					
Zaszumienie 50% (T)					
Zaszumienie 70% (T)					
Obraz idealny przysłonięty (T)					

Tabela 2. Zbiór obrazów odtworzonych rozmiar 8x10 pikseli

Obrazy inicjujące	Obrazy odtworzone											
	Zapis RP			Zapis RRD			Zapis RH			Zapis ZRP		
	Odczyt MAL	Odczyt MAC	Odczyt MS	Odczyt MAL	Odczyt MAC	Odczyt MS	Odczyt MAL	Odczyt MAC	Odczyt MS	Odczyt MAL	Odczyt MAC	Odczyt MS
												
												



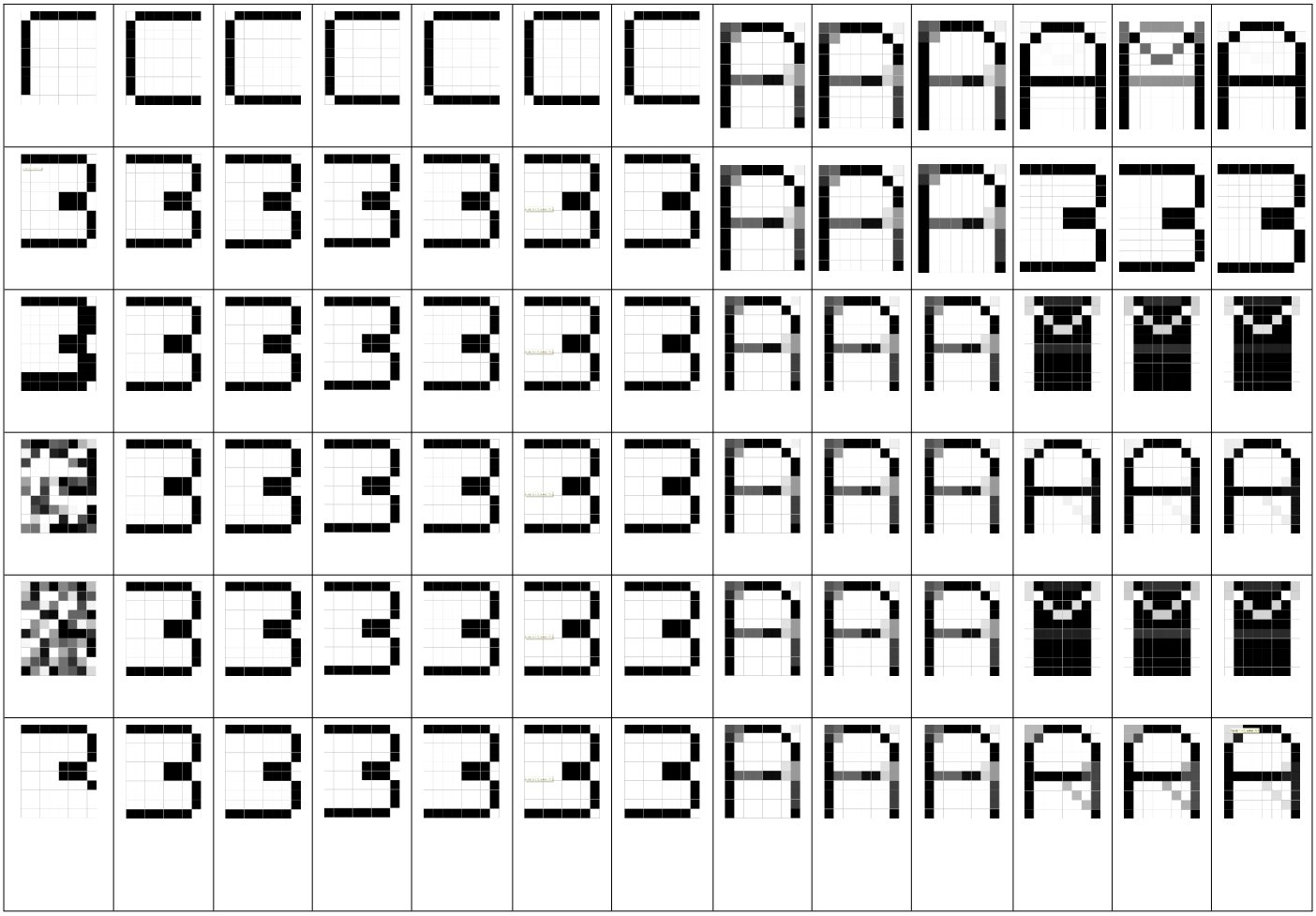
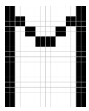
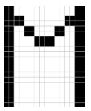
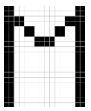
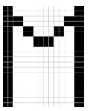
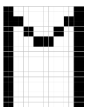
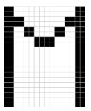
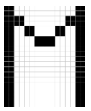
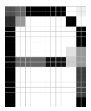
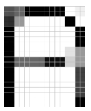
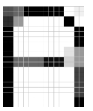
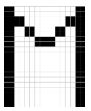
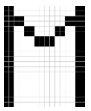
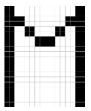
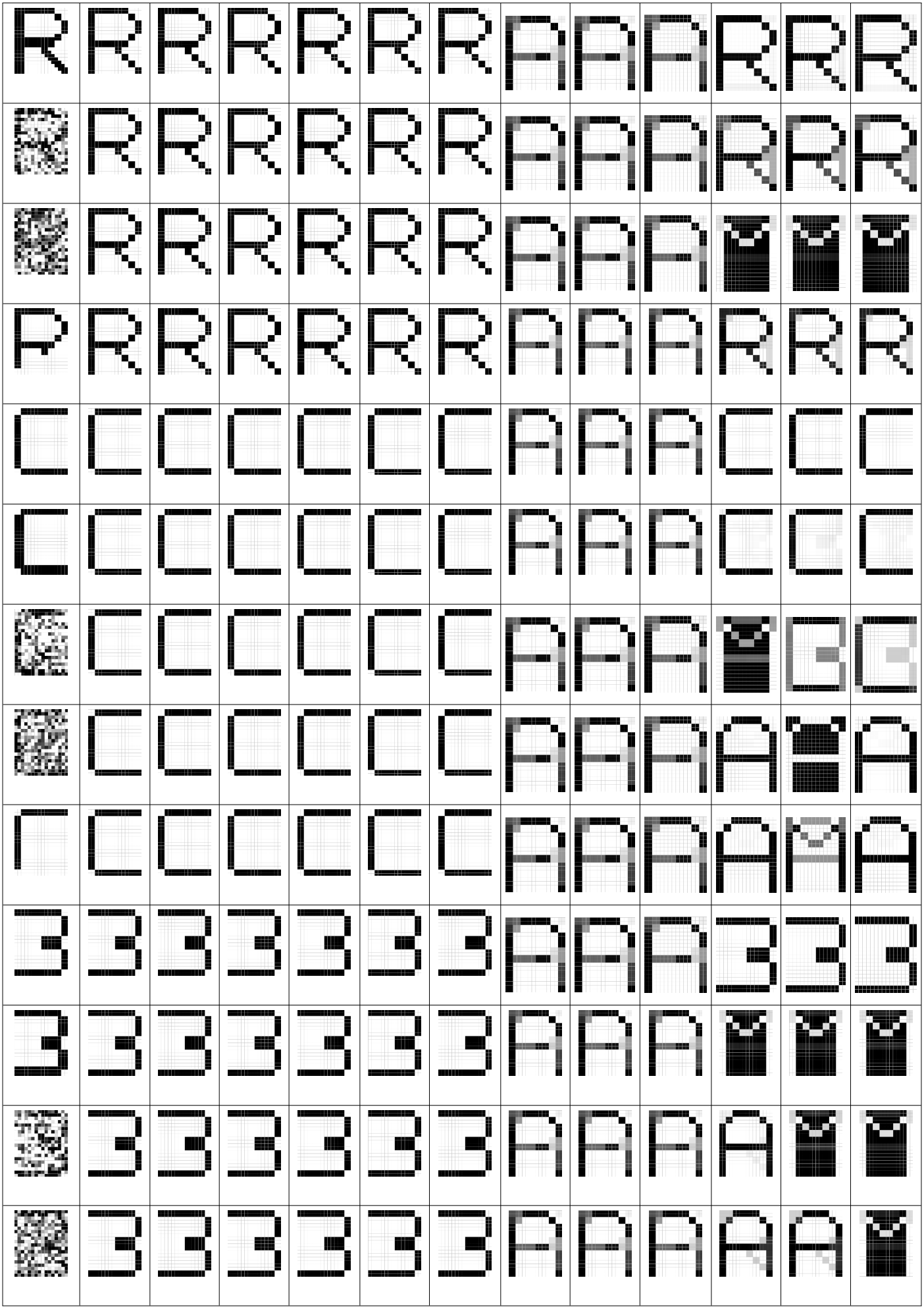
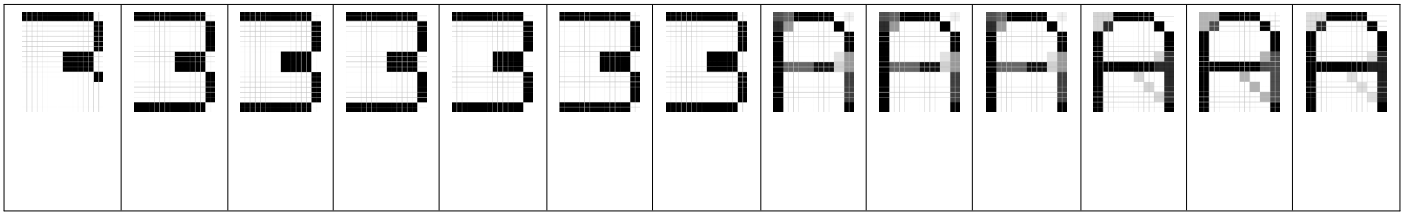


Tabela 3. Zbiór obrazów uczących i testowych 16x20 pikseli (U-uczące, T-testowe)

Litera	M	A	R	C	3
Obraz idealny (U)					
Zdeformowany obraz idealny (T)					
Zaszumienie 50% (T)					
Zaszumienie 70% (T)					
Obraz idealny przysłonięty (T)					

Tabela 4. Zbiór obrazów odtworzonych rozmiar 16x20 pikseli

Obrazy inicjujące	Obrazy odtworzone											
	Zapis RP			Zapis RRD			Zapis RH			Zapis ZRP		
	Odczyt MAL	Odczyt MAC	Odczyt MS	Odczyt MAL	Odczyt MAC	Odczyt MS	Odczyt MAL	Odczyt MAC	Odczyt MS	Odczyt MAL	Odczyt MAC	Odczyt MS